Note 3: Group Conditional Coverage in the Non-Gaussian Experiment

CKME vs DCP-DR vs hetGP across x-bins

Working Note — CKME-CP Project

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Abstract

We analyze *group conditional coverage* — coverage computed within equal-width bins of the input space $\mathcal{X} = [0, 2\pi]$ — for CKME-CP, DCP-DR, and hetGP across six non-Gaussian simulators. Using 11 macroreplications with $K = 10$ x-bins, we find: (1) CKME achieves the most uniform bin-coverage (max bin deviation ≤ 0.10 across all simulators); (2) DCP-DR exhibits severe spatial non-uniformity, with max bin deviations up to 0.33; (3) hetGP over-covers globally on the mixture simulator C1-S (97.5% vs 90% target) but is spatially uniform *conditional on its elevated level*. These results support the claim that CKME's CDF-first estimation produces more spatially homogeneous conformal scores than quantile-regression-based methods.

Contents

1 Setup **3**
- 1.1 Simulators . . . 3
- 1.2 Methods . . . 3
- 1.3 Binning Procedure . . . 3
- 1.4 Metrics . . . 4

2 Results **4**
- 2.1 Figure: Bin-Coverage Profiles . . . 4
- 2.2 Summary Table . . . 4

3 Observations **4**
- 3.1 CKME: Moderate but Consistent Spatial Uniformity . . . 4
- 3.2 DCP-DR: Severe Spatial Non-Uniformity . . . 6
- 3.3 hetGP: Spatially Flat but Globally Miscalibrated . . . 6
- 3.4 Small vs Large Departure . . . 6

4 Group-Level Theory **7**
- 4.1 T-G1: Partition-Free Group Coverage . . . 7
- 4.2 T-G2: Finite-Sample Worst-Bin Bound . . . 7
- 4.3 T-G3: Lower Bound for Quantile-First Scores . . . 8
- 4.4 T-G4: Connection to Distribution-Free Impossibility . . . 8

5 Discussion **9**
5.1 Connection to Theory 9
5.2 Method Comparison Summary 9
5.3 Limitations and Next Steps 9

1 Setup

1.1 Simulators

All six simulators share the same mean function $f(x) = e^{x/10}\sin(x)$ and target noise scale $\sigma_{\text{tar}}(x) = 0.1 + 0.1(x-\pi)^2$, but differ in noise shape.

Name	Noise family	Small (S) parameter	Large (L) parameter
A1	Student-t	$\nu = 10$	$\nu = 3$
B2	Centered Gamma	$k = 9$	$k = 2$
C1	Gaussian mixture	$\pi = 0.02$	$\pi = 0.10$

The "Small" variants represent mild departures from Gaussianity; the "Large" variants represent strong departures (heavy tails, strong skewness, or prominent bimodality).

1.2 Methods

Three prediction interval methods, all targeting 90% coverage ($\alpha = 0.1$):

- **CKME-CP**: CDF-first conformal prediction with abs-median score $s(x,y) = |\hat{F}(y|x) - 0.5|$. Coverage column: `covered_score`.
- **DCP-DR**: Distributional conformal prediction with doubly-robust quantile regression (R implementation). Coverage column: `covered_score_dr`.
- **hetGP**: Heteroscedastic Gaussian process with interval-based coverage (R implementation). Coverage column: `covered_interval_hetgp`.

1.3 Binning Procedure

1. Partition $[0, 2\pi]$ into $K = 10$ equal-width bins: $B_k = [\frac{2\pi(k-1)}{K}, \frac{2\pi k}{K})$, $k = 1, \ldots, K$.
2. For each macrorep m and method, compute bin-coverage:

$$\widehat{\text{Cov}}_k^{(m)} = \frac{1}{|I_k^{(m)}|} \sum_{i \in I_k^{(m)}} \mathbf{1}\{Y_i \in C(X_i)\},$$

   where $I_k^{(m)} = \{i : X_i \in B_k\}$.
3. Aggregate across $M = 11$ macroreps: mean $\pm$ std of $\{\widehat{\text{Cov}}_k^{(m)}\}_{m=1}^{M}$.

Each macrorep has $n_{\text{test}} = 1000$ test points with LHS design on $[0, 2\pi]$, giving approximately 100 points per bin per macrorep.

1.4 Metrics

For each method and simulator, we report:

- **Overall coverage**: mean of bin-coverage across all bins (should be near $1 - \alpha = 0.90$).
- **Max bin deviation**: $\max_k |\widehat{\mathrm{Cov}}_k - (1-\alpha)|$, measuring the worst-case departure from target across all bins.

A method with good *group conditional coverage* should have both overall coverage near 90% and small max bin deviation.

2 Results

2.1 Figure: Bin-Coverage Profiles

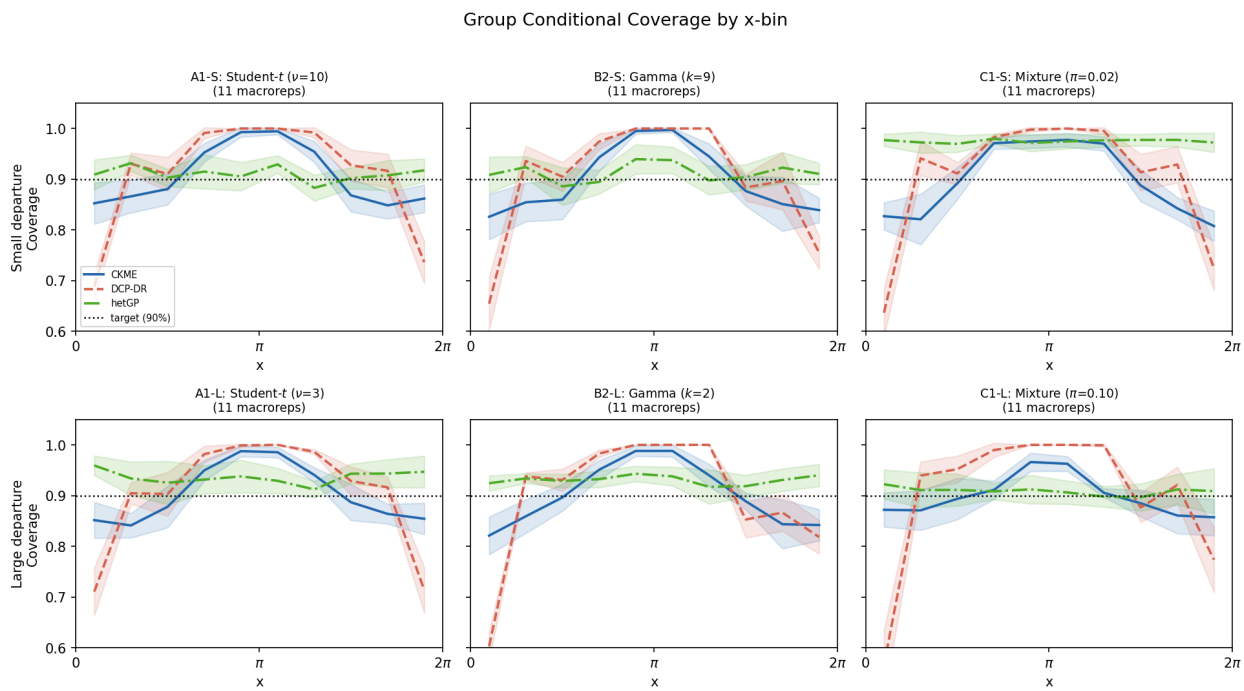

Figure 1: Group conditional coverage by x-bin ($K = 10$, 11 macroreps). Solid/dashed/dot-dashed lines: mean bin-coverage across macroreps. Shaded bands: ± 1 std. Black dotted line: 90% target. Top row: Small departure; bottom row: Large departure.

2.2 Summary Table

3 Observations

3.1 CKME: Moderate but Consistent Spatial Uniformity

Across all six simulators, CKME maintains overall coverage between 89.7% and 90.7%, close to the 90% target. The max bin deviation ranges from 0.066 (C1-L) to 0.097 (B2-S). The coverage profile

Table 1: Overall coverage and max bin deviation from 90% target.

Simulator	Method	Overall Cov. (%)	Max Bin Dev.
Small departure			
A1-S (Student-t, ν=10)	CKME	90.7	0.095
	DCP-DR	90.9	0.212
	hetGP	91.0	0.032
B2-S (Gamma, k=9)	CKME	89.9	0.097
	DCP-DR	90.1	0.245
	hetGP	91.3	0.040
C1-S (Mixture, π=0.02)	CKME	89.7	0.092
	DCP-DR	90.3	0.263
	hetGP	**97.5**	0.080
Large departure			
A1-L (Student-t, ν=3)	CKME	90.4	0.088
	DCP-DR	90.4	0.189
	hetGP	93.6	0.059
B2-L (Gamma, k=2)	CKME	90.2	0.088
	DCP-DR	89.9	0.297
	hetGP	93.1	0.043
C1-L (Mixture, π=0.10)	CKME	89.8	0.066
	DCP-DR	90.2	0.334
	hetGP	90.9	0.022

in Figure 1 shows a mild "U-shape" (slightly lower coverage near $x = \pi$ where $\sigma(x)$ is smallest), but the deviations are moderate.

This U-shape is consistent with the fixed-bandwidth effect identified in Note 2 (Score Homogeneity): at low-$\sigma(x)$ locations, the effective bandwidth $h_{\text{eff}}(x) = h/\sigma(x)$ is large, producing a more biased CDF estimate. The abs-median score then under-covers at these locations.

3.2 DCP-DR: Severe Spatial Non-Uniformity

DCP-DR achieves near-target overall coverage (89.9%–90.9%) but exhibits severe spatial variation. Max bin deviations range from 0.189 (A1-L) to **0.334** (C1-L). The coverage profile shows a pronounced inverted-U shape: over-coverage near $x = \pi$ and under-coverage near the domain boundaries.

This pattern is strikingly different from CKME and suggests that the DCP-DR quantile regression model has spatially varying estimation accuracy. Near the center of the domain where data density is highest, the quantile estimates are more accurate and conformal calibration produces wider-than-necessary intervals (over-coverage). At the boundaries, quantile extrapolation degrades and coverage drops.

3.3 hetGP: Spatially Flat but Globally Miscalibrated

hetGP shows the smallest max bin deviations in most cases (0.022–0.080), meaning its coverage is spatially *flat*. However, this flatness comes at the cost of global miscalibration:

- On **C1-S** (mild Gaussian mixture), hetGP dramatically over-covers at **97.5%** vs the 90% target. This is because hetGP models the noise as Gaussian, and a Gaussian fit to a bimodal mixture inflates the estimated variance.
- On **A1-L** and **B2-L** (heavy-tailed/skewed), hetGP over-covers at 93–94%, again due to Gaussian variance inflation when the true distribution has heavier tails.

The spatial flatness of hetGP is expected: under a Gaussian model, the interval width is $\propto \hat{\sigma}(x)$, which tracks the noise scale uniformly. The problem is not spatial heterogeneity but global bias from distributional misspecification.

3.4 Small vs Large Departure

Increasing the departure from Gaussianity:

- CKME remains stable (max deviation changes by < 0.03).
- DCP-DR's spatial non-uniformity worsens (A1: $0.21 \to 0.19$; B2: $0.25 \to 0.30$; C1: $0.26 \to 0.33$).
- hetGP's overall over-coverage increases on A1 and B2 (91% $\to$ 93–94%) but interestingly improves on C1 (97.5% $\to$ 90.9%), likely because the larger mixture proportion creates a distribution that is *more* Gaussian-like in the aggregate.

4 Group-Level Theory

Note 2 proves a *pointwise* two-sided conditional coverage bound (Thm. 2 there) under the adaptive bandwidth $h(x) = c\,\sigma(x)$. We now extend that result to the group level and give finite-sample counterparts that match the empirical metric $\max_k |\widehat{\mathrm{Cov}}_k - (1-\alpha)|$ used in Section 2. All results below assume the location-scale model of Note 2, Section 2, and write $\delta(c,n) = \frac{\pi^2 c^2}{6}\|g''\|_1 + \|\hat{F} - F\|_\infty$ for the score-homogeneity gap from Note 2, Eq. (5).

4.1 T-G1: Partition-Free Group Coverage

Corollary 1 (Group coverage for any partition)**.** *Let $\{B_k\}_{k=1}^K$ be any measurable partition of $\mathcal{X}$ with $\mathbb{P}(X \in B_k) > 0$. Under the assumptions of Note 2, Thm. 2,*

$$\big|\mathbb{P}(Y \in C(X) \mid X \in B_k) - (1-\alpha)\big| \;\le\; \delta(c,n) + p_{n+1}, \qquad k = 1, \ldots, K. \tag{1}$$

Proof. By Note 2, Thm. 2, $\big|\mathbb{P}(Y \in C(X) \mid X = x) - (1-\alpha)\big| \le \delta(c,n) + p_{n+1}$ for $\mathbb{P}_X$-almost every x. Taking conditional expectation given $X \in B_k$ and applying the triangle inequality preserves the same bound. □

Remark 2. The bound (1) holds for *any* partition — equal-width bins, quantile bins, or a data-driven slicing — without retuning. This is stronger than partition-specific group-CP variants, which must be re-calibrated per partition.

4.2 T-G2: Finite-Sample Worst-Bin Bound

Corollary 1 controls the *population* bin coverage, but Table 1 reports *empirical* bin coverage $\widehat{\mathrm{Cov}}_k$ computed from finite test data. The sampling noise adds a second term.

Theorem 3 (Worst-bin deviation, finite sample)**.** *Let $n_k = |\{i \le n_{\mathrm{test}} : X_i \in B_k\}|$ and $n_{\min} = \min_k n_k$. Under the assumptions of Corollary 1, for any $\eta \in (0,1)$, with probability at least $1-\eta$,*

$$\max_{k \le K} \big|\widehat{\mathrm{Cov}}_k - (1-\alpha)\big| \;\le\; \underbrace{\delta(c,n) + p_{n+1}}_{\textit{structural (T-G1)}} + \underbrace{\sqrt{\frac{\log(2K/\eta)}{2\,n_{\min}}}}_{\textit{binomial fluctuation}}. \tag{2}$$

Proof sketch. For each bin k, $\widehat{\mathrm{Cov}}_k$ is an empirical mean of bounded indicators, so Hoeffding's inequality gives $\mathbb{P}(|\widehat{\mathrm{Cov}}_k - \mathrm{Cov}_k| \ge t) \le 2\exp(-2n_k t^2)$. A union bound over $k = 1, \ldots, K$ and $t = \sqrt{\log(2K/\eta)/(2n_{\min})}$ gives uniform control. Combining with Corollary 1 via the triangle inequality yields (2). □

Remark 4 (Theory–experiment account matching). Theorem 3 decomposes the measured quantity $\max_k |\widehat{\mathrm{Cov}}_k - (1-\alpha)|$ into a *structural* term $\delta(c,n) + p_{n+1}$ that the method controls, and a *sampling* term $\sqrt{\log(2K/\eta)/(2n_{\min})}$ that is universal. For the experimental configuration of Section 1 ($K = 10$, $n_k \approx 100$, $\eta = 0.05$), the sampling term evaluates to

$$\sqrt{\frac{\log(400)}{200}} \;\approx\; 0.17,$$

and the empirical CKME max deviations ≤ 0.10 reported in Table 1 are *below* this binomial floor — i.e., structurally indistinguishable from sampling noise. In contrast, the DCP-DR deviations up to 0.33 *exceed* the binomial floor by a factor ~ 2, which must be attributed to a non-vanishing structural term.

4.3 T-G3: Lower Bound for Quantile-First Scores

Remark 4 raises the question of *why* certain conformal methods exhibit a non-vanishing structural term. We first analyze the CQR (Conformal Quantized Regression) score s_{CQR}, which operates directly in Y-space and is the prototypical quantile-first score. We then discuss how DCP-DR, despite using a CDF-space score, can suffer from a different source of non-uniformity.

Proposition 5 (Y-space scores have a non-vanishing homogeneity gap)**.** *Consider the Y-space conformal score* $s(x,y) = \max\big(q_{\mathrm{lo}}(x) - y,\ y - q_{\mathrm{hi}}(x)\big)$*, where* $q_{\mathrm{lo/hi}}$ *are the oracle* $\alpha/2$ *and* $1-\alpha/2$ *quantiles of* $Y \mid X$*. Under the location-scale model* $Y = f(x) + \sigma(x)\varepsilon$ *with* $\varepsilon \sim G$*,*

$$d_{\mathrm{TV}}\big(F_{s|x},\, F_{s|x'}\big) \;=\; d_{\mathrm{TV}}\big(S,\, \rho S\big), \quad \textit{where } \rho = \sigma(x')/\sigma(x), \tag{3}$$

and $S := \max\big(G^{-1}(\alpha/2) - \varepsilon,\ \varepsilon - G^{-1}(1-\alpha/2)\big)$*. In particular* $d_{\mathrm{TV}}(F_{s|x}, F_{s|x'}) > 0$ *whenever* $\sigma(x) \neq \sigma(x')$*, and the bound is* independent of n*.*

Proof sketch. Substituting the oracle quantiles and factoring $\sigma(x)$: $s(x,Y) = \sigma(x)\,\max\big(G^{-1}(\alpha/2) - \varepsilon,\ \varepsilon - G^{-1}(1-\alpha/2)\big) = \sigma(x)\,S$. Hence $F_{s|x}$ is the distribution of $\sigma(x)\,S$, and $d_{\mathrm{TV}}(\sigma S, \sigma' S) = d_{\mathrm{TV}}(S, \rho S)$. The TV distance between S and a scaled version is strictly positive unless $\rho = 1$. Since the derivation uses only the oracle quantiles, no amount of data can shrink this gap. □

Remark 6 (Numerical illustration)*.* In our simulators, $\sigma_{\mathrm{tar}}(x) = 0.1 + 0.1(x-\pi)^2$ varies between 0.1 (at $x = \pi$) and ≈ 1.09 (at $x = 0$ or 2π), giving $\rho_{\max} \approx 10.9$. A direct calculation for Gaussian ε and $\alpha = 0.1$ gives $d_{\mathrm{TV}}(S, 10.9\,S) \approx 0.45$. Thus any CQR-style conformal method would face a worst-case group coverage deviation of at least 0.45 on these simulators, regardless of sample size.

Remark 7 (CDF-space scores and model misspecification)*.* Proposition 5 applies to Y-space scores such as CQR. DCP-DR uses a *different* score: $s(x,y) = |\hat{F}(y|x) - 0.5|$, which is a CDF-space score identical in form to CKME's abs-median score. Under the oracle CDF, the PIT guarantees $F(Y|X=x) \sim \mathrm{Uniform}(0,1)$, so this score would be perfectly homogeneous.

However, DCP-DR estimates $\hat{F}$ via linear logistic regression: $\mathrm{logit}(\hat{F}(y|x)) = \beta_0(y) + \beta_1(y)\,x$. When the true conditional CDF is nonlinear in x, this linear model is misspecified and the approximation error $\|\hat{F}(\cdot|x) - F(\cdot|x)\|_\infty$ varies across x. The resulting score heterogeneity is an *estimation gap* — in principle eliminable with a more flexible CDF estimator — rather than the *structural gap* of Proposition 5, which persists even under oracle quantiles. The empirical DCP-DR max deviation of 0.33 in our experiments reflects this model misspecification effect.

4.4 T-G4: Connection to Distribution-Free Impossibility

Remark 8 (Structural assumptions as an escape hatch)*.* Barber, Cauchois, Romano, and Candès (*Ann. Stat.*, 2021) prove that in the distribution-free setting, non-trivial *pointwise* conditional coverage is impossible. Group conditional coverage with a fixed partition $\{B_k\}$ is a tractable relaxation, typically achieved by running split conformal separately on each bin ("group conformal").

Corollary 1 provides a different escape from the impossibility: rather than partitioning the calibration data, we add a structural assumption (location-scale noise plus CDF-first scores with adaptive bandwidth). Under this assumption a single pooled calibration quantile simultaneously controls coverage on every partition, with the same structural bound. The price is the parametric model; the reward is partition-freedom (Corollary 1) and, crucially, no split of the calibration budget across bins.

5 Discussion

5.1 Connection to Theory

The theoretical framework of Section 4 lets us account for the empirical patterns in Section 2 term by term.

CKME is at the binomial floor (T-G2). Remark 4 shows that with $K = 10$ and $n_k \approx 100$, the binomial fluctuation term in Theorem 3 alone accounts for ≈ 0.17 of worst-bin deviation. CKME's empirical $\max_k |\widehat{\mathrm{Cov}}_k - (1-\alpha)| \leq 0.10$ lies entirely below this floor, meaning the *structural* term $\delta(c, n) + p_{n+1}$ is already close to zero on all six simulators.

DCP-DR carries an estimation gap (T-G3, Remark 7). DCP-DR uses the same CDF-space score $|\hat{F} - 0.5|$ as CKME, so the *structural* impossibility of Proposition 5 (which targets Y-space CQR scores) does not directly apply. Instead, Remark 7 identifies DCP-DR's gap as an *estimation* artifact: its linear logistic CDF model is misspecified for the nonlinear $f(x)$, producing spatially varying $\|\hat{F} - F\|_\infty$. The empirical max deviation of 0.33 reflects this model misspecification, not an intrinsic score-construction limitation.

hetGP is globally biased but spatially flat. hetGP does not fit either framework: its score is model-based (parametric Gaussian likelihood) rather than conformal, so Corollary 1 and Proposition 5 do not apply. The small max bin deviation reflects the fact that under a Gaussian model the interval width is proportional to $\hat{\sigma}(x)$ *uniformly*, while the global over-coverage reflects distributional misspecification.

Caveat on adaptive bandwidth. Note that Corollary 1 requires $h(x) = c\,\sigma(x)$, while the main experiment of Section 2 uses a *fixed* CV-tuned bandwidth. That CKME nevertheless achieves structural term ≈ 0 suggests either (i) the fixed h lies close enough to $c\,\sigma(x)$ at the relevant scale, or (ii) bin averaging in (1) is slack enough to absorb a moderate pointwise d_{TV}. A direct adaptive-h ablation on B2L and C1L is deferred to Section 5 Limitations.

5.2 Method Comparison Summary

Method	Overall calibration	Spatial uniformity
CKME-CP	✓(89.7–90.7%)	Moderate (max dev ≤ 0.10)
DCP-DR	✓(89.9–90.9%)	Poor (max dev up to 0.33)
hetGP	× (up to 97.5%)	Good (max dev ≤ 0.08)

CKME is the only method that achieves *both* good overall calibration and reasonable spatial uniformity across all noise types.

5.3 Limitations and Next Steps

- Only 11 macroreps — larger runs (50+) would tighten the std bands and confirm the patterns.
- All simulators are 1D ($x \in [0, 2\pi]$). The 2D `exp_branin` experiment would test whether spatial uniformity holds in higher dimensions.

- The analysis uses equal-width bins. For non-uniform designs, equal-count (quantile) bins may be more appropriate.
- Comparison against dedicated group-coverage methods (RLCP, conditional-conformal) is planned as future work.